

Architectural Enhancement: Residual Connections for Improved Gradient Flow in PyramidAttnCNN

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January 8, 2026

1 Introduction and Motivation

While the baseline PyramidAttnCNN architecture achieved strong performance on IMU-based joint biomechanics prediction (approximately 3.0% normalized RMSE for angles and 3.5% for moments), several opportunities for architectural enhancement remained. The baseline model employed three sequential convolutional layers per branch with progressive channel expansion ($24 \rightarrow 48 \rightarrow 96$ channels), creating a hierarchical feature learning pathway. However, this sequential stacking of convolutional layers, while effective for building increasingly abstract representations, can suffer from gradient flow issues during backpropagation, particularly in the early stages of training when weights are randomly initialized and gradient magnitudes can diminish as they propagate backward through multiple nonlinear activation functions.

The motivation for introducing residual connections stems from three key observations about the baseline architecture. First, even with only three convolutional layers per branch, the gradient signal must traverse multiple transformation stages (convolution, batch normalization, ReLU activation, dropout) at each layer, potentially leading to gradient attenuation. Second, the optimization landscape for learning complete feature transformations at each layer (mapping input features directly to output features) can be challenging, particularly when the optimal transformation is close to an identity mapping. Third, the extremely low dropout rates discovered through hyperparameter optimization (0.026 for convolutional layers) suggested that the model was not suffering from severe overfitting, indicating that the architecture could potentially benefit from increased representational capacity enabled by improved gradient flow rather than requiring stronger regularization.

Residual connections, introduced by He et al. in their seminal ResNet paper, address these challenges by reformulating layer learning as residual function learning. Instead of learning a direct mapping $\mathcal{H}(x)$, residual blocks learn a residual function $\mathcal{F}(x) = \mathcal{H}(x) - x$, with the final output computed as $\mathcal{H}(x) = \mathcal{F}(x) + x$. This reformulation provides several theoretical and practical advantages: it creates direct gradient pathways through skip connections that bypass nonlinear transformations, it simplifies the optimization problem by allowing the network to learn refinements rather than complete transformations, and it provides implicit regularization by constraining the solution space.

2 Residual Block Implementation

2.1 Mathematical Formulation

The core enhancement to the PyramidAttnCNN architecture involves replacing the standard convolutional layers in each branch with residual blocks. A standard convolutional layer in the baseline architecture computed:

$$y = \text{Dropout}(\text{ReLU}(\text{BatchNorm}(\text{Conv}(x)))) \quad (1)$$

The enhanced residual block reformulates this as:

$$y = \mathcal{F}(x, \{W_i\}) + \mathcal{S}(x) \quad (2)$$

where $\mathcal{F}(x, \{W_i\})$ represents the residual function (the main computational path) and $\mathcal{S}(x)$ represents the skip connection. The residual function is computed as:

$$\mathcal{F}(x) = \text{Dropout}(\text{ReLU}(\text{BatchNorm}(\text{Conv}(x)))) \quad (3)$$

For the skip connection $\mathcal{S}(x)$, we employ an adaptive strategy that depends on whether the input and output channel dimensions match:

$$\mathcal{S}(x) = \begin{cases} x & \text{if } C_{\text{in}} = C_{\text{out}} \\ \text{Conv}_{1 \times 1}(x) & \text{if } C_{\text{in}} \neq C_{\text{out}} \end{cases} \quad (4)$$

This adaptive approach is critical because in the PyramidAttnCNN architecture, channel dimensions change at each layer ($42 \rightarrow 24 \rightarrow 48 \rightarrow 96$), requiring projection of the skip connection through a 1×1 convolution to match dimensions for the element-wise addition. The 1×1 convolution provides a learnable linear projection that adapts the channel dimensionality while preserving the temporal structure.

2.2 Implementation Details

The residual block implementation consists of several carefully designed components. The main computational path follows the pre-activation residual architecture pattern, though with post-activation for compatibility with the baseline architecture. The convolutional layer maintains the same padding strategy as the baseline ($\text{padding} = \text{kernel_size} // 2$) to preserve temporal dimensions at 101 time points through the initial feature extraction layers.

The skip connection implementation requires special attention to channel dimension mismatches. In the PyramidAttnCNN architecture, the initial convolutional layer transforms from 42 input channels (IMU sensor data) to 24 channels, the second layer expands to 48 channels, and the third layer expands to 96 channels. Each of these transitions requires a 1×1 projection convolution for the skip connection:

- Layer 1: $42 \rightarrow 24$ channels (requires 1×1 conv for skip)
- Layer 2: $24 \rightarrow 48$ channels (requires 1×1 conv for skip)
- Layer 3: $48 \rightarrow 96$ channels (requires 1×1 conv for skip)

The 1×1 convolutional layers are initialized using PyTorch’s default initialization scheme and do not include batch normalization or activation functions, providing a pure linear projection for the skip connection.

3 Integration with PyramidAttnCNN Architecture

3.1 Selective Application of Residual Connections

A critical design decision involves determining where in the architecture to apply residual connections. The enhanced architecture applies residual blocks selectively to the hierarchical feature extraction layers while maintaining the original structure for downsampling and attention components. Specifically:

Residual connections applied to:

- Three hierarchical convolutional layers in each branch (those operating at full 101-timepoint resolution)
- Both branches (kernel sizes 3 and 7) receive identical residual enhancement

Residual connections NOT applied to:

- Aggressive downsampling layers (stride-4 and stride-3 convolutions)
- 1×1 dimensionality reduction layer after branch concatenation
- Multi-head attention mechanism
- Final prediction heads

This selective application reflects a deliberate architectural philosophy. The hierarchical feature extraction layers, operating at full temporal resolution with complex channel transformations, benefit most from improved gradient flow. The downsampling layers perform a fundamentally different function (temporal compression rather than feature learning) and do not require skip connections. The attention mechanism already operates on compressed representations and employs a different computational paradigm (attention-based aggregation) that does not directly benefit from residual connections in the same way.

3.2 Gradient Flow Analysis

The primary benefit of residual connections manifests in improved gradient flow during backpropagation. Consider the gradient computation through a standard three-layer convolutional stack in the baseline architecture:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial y_3} \cdot \frac{\partial y_3}{\partial y_2} \cdot \frac{\partial y_2}{\partial y_1} \cdot \frac{\partial y_1}{\partial x} \quad (5)$$

Each term in this chain involves derivatives through convolution, batch normalization, ReLU activation, and dropout operations. The ReLU activation in particular introduces gradient sparsity (gradients are exactly zero for negative inputs), and the multiplicative nature of the chain rule means that gradient magnitudes can decay exponentially as the number of layers increases.

With residual connections, the gradient computation gains an additional direct pathway:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial y} \cdot \left(\frac{\partial \mathcal{F}(x)}{\partial x} + \frac{\partial \mathcal{S}(x)}{\partial x} \right) \quad (6)$$

The skip connection term $\frac{\partial \mathcal{S}(x)}{\partial x}$ provides a direct gradient pathway. For identity skip connections, $\frac{\partial \mathcal{S}(x)}{\partial x} = I$ (the identity matrix), guaranteeing that at least some gradient signal flows directly backward without attenuation.

For projected skip connections (1×1 convolutions), $\frac{\partial \mathcal{S}(x)}{\partial x} = W_{1 \times 1}$, still providing a relatively direct pathway compared to the multi-stage main path.

This dual-pathway gradient flow has profound implications for training dynamics. Early in training, when the main path weights may produce small gradients, the skip connection ensures adequate gradient signal reaches earlier layers. As training progresses, the network can learn to balance contributions from both pathways, potentially using the skip connection as a baseline and the main path for refinements.

3.3 Information Preservation Through Skip Connections

A fundamental advantage of residual connections lies in their ability to preserve original information throughout the hierarchical transformation process. In standard feedforward architectures, each layer completely replaces its input with a transformed representation. Consider the baseline architecture’s three-layer processing pipeline: raw IMU sensor data undergoes transformation to extract low-level temporal patterns, these patterns are transformed again to capture mid-level coordination features, and finally these are transformed once more to produce high-level biomechanical representations. By the third layer, the original raw sensor signals—including potentially critical fine-grained details such as the precise shape of acceleration spikes during heel strike or high-frequency impact vibrations—have been abstracted away through multiple nonlinear transformations.

Residual connections fundamentally change this information flow paradigm by enabling additive rather than replacement-based transformations. At each residual block, the output comprises both the transformed features and the original input: $y = \mathcal{F}(x) + x$. This addition operation ensures that information from the input is directly propagated to the output alongside the learned transformation. Critically, this is element-wise addition, not concatenation, maintaining constant dimensionality while combining information streams. Through successive residual blocks, the final layer’s representation contains components traceable back through all previous layers to the original input, creating a cumulative feature hierarchy where early-layer patterns remain accessible even after multiple transformations.

For the biomechanics prediction task, this information preservation has concrete implications. Raw IMU accelerometer signals contain sharp transient features—such as the characteristic spike during heel strike that occurs over just a few milliseconds—that may correlate strongly with peak knee moments. In a standard architecture, these sharp features might be smoothed or abstracted away by the second or third convolutional layer’s learned fil-

ters, particularly if those layers optimize for different aspects of the signal. With residual connections, even if layer two and layer three learn to extract more abstract temporal coordination patterns, the original sharp transient features remain present in the residual pathway, available to subsequent layers and ultimately to the prediction heads. The network effectively learns both what to transform about the input and what to preserve unchanged, with the skip connections providing a mechanism for retaining discriminative raw features throughout the processing hierarchy.

The dimension matching challenge arises because the PyramidAttnCNN architecture progressively expands channel dimensions through its hierarchical layers, from forty-two input channels to twenty-four, then forty-eight, then ninety-six channels. Direct addition $y = \mathcal{F}(x) + x$ requires matching dimensions, yet the transformation changes channel count. The solution employs 1×1 convolutional projections for the skip connection when dimensions mismatch. Mathematically, for input $x \in \mathbb{R}^{B \times C_{\text{in}} \times T}$ and transformed output $\mathcal{F}(x) \in \mathbb{R}^{B \times C_{\text{out}} \times T}$ where $C_{\text{in}} \neq C_{\text{out}}$, the skip connection applies a learned linear projection $W_{\text{proj}} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$ such that $\mathcal{S}(x) = W_{\text{proj}}x$, yielding compatible dimensions for the addition operation. This projection is a 1×1 convolution that learns how to map the input channel space to the output channel space, effectively learning which combinations of input channels should be preserved and passed forward to match the expanded feature representation. Importantly, this projection uses no activation function, providing a pure linear mapping that preserves information content while adapting dimensionality. The resulting residual block computes $y = \mathcal{F}(x) + W_{\text{proj}}x$, where the original information passes through the learned projection while the transformation provides complementary learned features, and their sum forms the block output that feeds into subsequent layers.

3.4 Optimization Landscape Implications

Beyond gradient flow and information preservation, residual connections fundamentally alter the optimization landscape by reformulating the learning problem. Standard layers must learn complete feature transformations from scratch. If the optimal transformation for a particular layer is close to identity (i.e., the best thing to do is approximately $y \approx x$ with minor refinements), a standard layer must learn to approximate the identity function through its convolutional weights, which is non-trivial, especially after ReLU activation that introduces nonlinearity.

Residual blocks explicitly decompose this into identity plus residual: $y = x + \mathcal{F}(x)$. If identity is optimal, the residual block can simply learn $\mathcal{F}(x) \approx 0$,

which is much easier—the network can initialize with small weights and learn small refinements. This reformulation effectively makes the optimization problem easier, particularly in early training phases, potentially leading to faster convergence and better final solutions.

For the biomechanics prediction task, this property is particularly relevant during the hierarchical feature learning stages. The first convolutional layer extracts basic patterns from raw IMU sensor signals. The second layer might find that many of these patterns are already informative and need only minor refinement rather than complete transformation. The residual formulation allows the network to preserve useful first-layer features while learning incremental improvements, rather than forcing a complete re-encoding.

4 Expected Benefits and Theoretical Justification

4.1 Training Stability and Convergence

Residual connections are expected to improve training stability through multiple mechanisms. The direct gradient pathways reduce the risk of vanishing gradients that can cause training to plateau prematurely. The reformulated optimization problem should exhibit smoother loss landscapes with fewer sharp local minima, as the network can more easily recover from poor initialization by relying on skip connections while gradually learning useful residual functions.

Empirically, ResNet architectures consistently demonstrate more stable training curves with less oscillation compared to their plain counterparts. For the PyramidAttnCNN application, this could manifest as smoother convergence during the 15-epoch training period, with more consistent improvement across epochs rather than the occasional plateaus or temporary increases in validation loss that can occur with standard architectures.

4.2 Representational Capacity

A subtle but important benefit concerns representational capacity. The addition of residual connections increases the model’s parameter count slightly (due to the 1×1 projection convolutions), but more importantly, it increases the effective capacity by enabling deeper or wider architectures without gradient flow penalties. While the current implementation maintains the same three-layer depth as the baseline, the improved gradient flow means the network can more effectively utilize the capacity of all three layers simultaneously.

In the baseline architecture, the third layer’s learning might be constrained by weak gradients flowing through layers two and one. With residual connections, all three layers receive stronger gradient signals, allowing them to learn complementary features more effectively. This could lead to richer, more discriminative feature representations that better capture the complex relationships between IMU sensor patterns and joint biomechanics.

4.3 Regularization and Generalization

Interestingly, residual connections provide implicit regularization effects. The skip connections constrain the solution space—the network’s output is always anchored to the input plus learned residuals, preventing the network from learning arbitrary transformations that might overfit to training data. This implicit regularization might partially explain why ResNet architectures often generalize well despite having more parameters than comparable plain networks.

For the biomechanics prediction task, this could translate to improved generalization across different participants, walking speeds, and environmental conditions. The model learns refinements to sensor patterns rather than memorizing specific training examples, potentially leading to better performance on the held-out test fold in cross-validation.

5 Experimental Considerations and Hypotheses

5.1 Expected Performance Improvements

Based on theoretical considerations and empirical evidence from residual architectures in other domains, we hypothesize several specific performance improvements:

Training dynamics: Faster convergence, potentially achieving baseline performance in fewer epochs (perhaps 10-12 epochs instead of 15). Smoother loss curves with reduced oscillation in both training and validation metrics.

Final performance: Modest improvement in test set normalized RMSE, potentially 0.1-0.3 percentage points (e.g., from 3.0% to 2.7-2.9% for angles). Larger improvements are possible but not guaranteed, as the baseline already achieves strong performance.

Generalization: Reduced train-test gap, indicating improved generalization. The correlation gap might decrease from approximately 0.05-0.10 to 0.03-0.08, and the nRMSE gap might decrease proportionally.

Peak moment prediction: Potentially larger improvements for peak moment prediction specifically, as this is often the most challenging aspect of the task and might benefit most from improved feature learning in early layers.

5.2 Comparison Methodology

To rigorously evaluate the residual enhancement, the experimental protocol maintains all hyperparameters identical to the baseline: same learning rate (0.00479), same batch size (12), same dropout rates (0.026 conv, 0.0024 FC), same Gaussian noise augmentation (0.1), same weight decay ($3.2\text{e-}5$), same gradient clipping (2.65), and same training duration (15 epochs). The only architectural difference is the replacement of standard convolutional layers with residual blocks in the hierarchical feature extraction stage.

Both models are trained using the same 5-fold cross-validation scheme with identical data splits, ensuring fair comparison. The multi-seed ensemble approach (three models per fold with different random seeds) provides robust performance estimates with uncertainty quantification. Metrics for comparison include normalized RMSE, correlation coefficients, standard deviation ratios, and train-test generalization gaps across all folds and axes.

5.3 Potential Limitations

Despite the theoretical benefits, several factors might limit the practical improvements from residual connections. First, the baseline architecture already achieves strong performance, leaving limited room for improvement. Second, the architecture is relatively shallow (three layers), and residual connections provide diminishing benefits in very shallow networks compared to deeper architectures (10-50+ layers) where gradient flow issues are more severe. Third, the extremely low optimal dropout rates suggest the baseline is not suffering from severe optimization difficulties, potentially indicating that gradient flow was already adequate.

Additionally, the biomechanics prediction task has inherent noise and variability—IMU sensor measurements contain measurement error, participant movement exhibits natural variability, and the gold-standard optical motion capture ground truth itself has uncertainty. These sources of irreducible error place a floor on achievable performance that architectural improvements cannot overcome.

6 Conclusion

The integration of residual connections into the PyramidAttnCNN architecture represents a principled enhancement based on solid theoretical foundations and extensive empirical evidence from the broader deep learning literature. By reformulating hierarchical feature learning as residual function learning and providing direct gradient pathways through skip connections, the enhanced architecture addresses potential optimization challenges while maintaining the successful multi-scale, attention-based design of the baseline.

The selective application of residual blocks to the hierarchical feature extraction layers reflects careful architectural reasoning—enhancing the components that benefit most from improved gradient flow while preserving the proven effectiveness of the attention mechanism and prediction heads. Whether these theoretical benefits translate to meaningful performance improvements on the specific task of IMU-based joint biomechanics prediction remains an empirical question to be answered through rigorous experimentation, but the architectural enhancement is well-motivated and implemented following best practices from the ResNet literature.

7 Next Steps and Future Directions

The most immediate priority involves empirical validation of the residual enhancement through direct performance comparison with the baseline architecture. Both models will be trained using identical hyperparameters and cross-validation schemes across all five folds, with the multi-seed ensemble approach providing robust performance estimates. Key metrics for comparison include normalized RMSE for joint angles and moments, correlation coefficients, standard deviation ratios, and train-test generalization gaps. Beyond aggregate metrics, we will analyze performance on specific challenging aspects of gait prediction, particularly peak knee moments during loading response, where improved gradient flow in early layers may provide the most substantial benefits. Training dynamics will be examined through loss curves, convergence rates, and gradient magnitude monitoring to verify whether the theoretical advantages of residual connections manifest empirically in this specific biomechanics prediction context.

Beyond immediate validation, the residual architecture opens several promising avenues for further enhancement. The improved gradient flow provided by skip connections enables exploration of deeper architectures

without encountering the degradation problem that would limit plain networks. A natural extension involves evaluating four-layer and five-layer variants to determine whether additional hierarchical feature learning stages capture more complex temporal patterns in IMU data that correlate with joint biomechanics. The current implementation uses post-activation residual blocks for compatibility with the baseline, but pre-activation variants, where batch normalization and ReLU precede convolution, have shown advantages in very deep networks and warrant investigation. Additionally, the residual framework could incorporate more sophisticated skip connection strategies such as dense connections, where each layer receives inputs from all preceding layers, or attention-weighted skip connections that learn to adaptively modulate the contribution of different pathways.

The architectural flexibility enabled by residual connections also facilitates investigation of alternative design choices within the PyramidAttnCNN framework. The current channel progression of twenty-four, forty-eight, and ninety-six channels represents one point in the design space, but residual connections might enable effective learning with wider channels at each stage or more aggressive expansion ratios. The two-branch multi-scale design with kernel sizes three and seven could be extended to three or four branches capturing an even broader range of temporal receptive fields, with residual connections ensuring that the increased depth does not impair gradient flow. Furthermore, the interaction between residual connections and the attention mechanism deserves deeper investigation—while the current implementation applies residual blocks only to the convolutional feature extraction stage, residual connections within or around the attention mechanism itself might provide additional benefits for learning which temporal phases are most predictive of joint biomechanics.

From a broader research perspective, the residual enhancement methodology demonstrated here generalizes beyond the specific task of knee joint prediction from IMU data. The same architectural principles could be applied to prediction of hip and ankle biomechanics, to different movement tasks beyond walking such as running or stair climbing, or to prediction of ground reaction forces and muscle activations. Transfer learning experiments could investigate whether residual models pre-trained on large datasets of healthy gait generalize more effectively to pathological populations than plain networks, potentially due to the more robust feature representations enabled by improved gradient flow. The residual architecture might also facilitate multi-task learning where a single model simultaneously predicts kinematics and kinetics across multiple joints, as the improved gradient flow could help prevent the gradual forgetting of early tasks that

sometimes occurs in multi-task settings. Finally, interpretability analysis through gradient-based attribution methods and attention weight visualization could provide insights into how residual connections change the learned representations and whether the skip connections primarily serve as gradient highways during training or continue to play an active role in the final trained model’s computations.